



Lab Assignment – 2.2 (SQL DDL and DML)

Department: CSE

Subject: DIS Lab
Code: 22CPP217

Objective: The objective of this experiment is to perform conditional data retrieval and manipulation using SQL DML commands. This assignment focuses on applying logical operators (AND, OR, NOT, BETWEEN, IN), filtering conditions, and arithmetic expressions. It also introduces transferring data between tables using INSERT INTO ... SELECT.

Reference : DML commands

Pre-Lab Setup : Run this block to clean your environment, disable "Safe Update" restrictions, and create the necessary tables.

SQL

```
-- 0. Disable Safe Update Mode (Prevents errors on bulk updates)
SET SQL_SAFE_UPDATES = 0;

-- 1. Reset Tables
DROP TABLE IF EXISTS table_name;

-- 2. Create Main Table <student_registry> with following column names
<ID, Name, Dept_Name, Total_Credits, Email>
-- Apply constraints-
    ID:primary key
    Name:not null

-- 3. Create Second Table for next question.
    <Cand_ID,Cand_Name, Cand_Dept, Cand_Credits>

-- 4. Insert Data from given link below
    Table Data Link
```

Q 1: The academic office needs to analyze student records based on strict criteria.

- Retrieve the ID, Name, and Credits of students whose total credits lie **between 40 and 80** (inclusive).
- Display the details of students who do not have an email address assigned (IS NULL).
- Retrieve all student records where the department name **starts with** the letter 'C' (e.g., Comp. Sci., Chemistry).
- Retrieve the Name and Department of students who belong to **either** 'History' **or** 'Physics'. Use the IN operator.
- Run a query to display a list of all unique departments currently in the registry (remove duplicates using DISTINCT).

Q 2: Apply complex filters to find specific groups of students.

- Display all students from the 'Comp. Sci.' department who have **more than 30 credits AND less than 100 credits**.
- Retrieve students who are **NOT** in the 'Comp. Sci.' department.
- Find students who either have **0 credits** OR belong to the 'General' department.

Q 3: A raw list of new candidates is available in the table **New_Candidates**. You must migrate them to **Student_Registry**, but you must filter out bad data during the transfer.

- Write a single SQL query to insert rows from **New_Candidates** into **Student_Registry** only if:
 1. The department is **NOT NULL**.
 2. The credits are **greater than or equal to 0**.
- Note that the column names in the two tables are different (Cand_ID vs ID). SQL maps them by **Order**, not by name.
- After running your query, **Valid Student** should be in the registry, but **Null Dept Student** and **Negative Creds** should be rejected.

Q 4: Perform bulk updates using arithmetic expressions and string functions.

- Increase the **Total_Credits** by **5** for all students in the 'Comp. Sci.' department who currently have less than 60 credits.
- For students who still have NULL emails, update their email column by concatenating their ID with '@univ.edu'.
 - **Hint:** CONCAT(ID, 'word').
- Increase the total credits by **2** for students whose current credit score is an **odd number**.
 - **Hint:** modulo operator: col_name % 2.

Q 5: Data Cleanup Remove invalid or obsolete records to maintain integrity.

- Delete any student whose department is 'General' **AND** has less than 10 credits.
- Delete any remaining records where the ***Total_Credits*** are negative (just in case bad data slipped in).